6. THE NUCLEAR REACTIONS

I. ALPHA RADIATION

Much has been said in the preceding lectures about the changes of atomic nuclei. In such changes, one chemical element turns into another, and in this respect modern nuclear physics has realized, to a certain extent, the hopes which inspired the alchemists of past ages. Let us now consider these nuclear transmutations more closely. The following two questions arise here: What elements can be changed into each other? Under what conditions is such a transmutation at all possible? In order to answer these questions, we will classify the transmutation processes into two groups: Firstly, the reactions which occur spontaneously, and secondly, those produced by external agencies.

The spontaneous change of an element is called *radioactivity,* for the process is accompanied by an emission of radiation. The radioactive elements can be further subdivided into two classes: Firstly, those which emit alpha radiation, and secondly, those which emit beta radiation (either electrons or positrons). Gamma radiation may occur simultaneously, too. In addition to these, there are a few other processes, which will be discussed later.

We shall begin with those transmutation processes in which alpha rays are emitted. Alpha radiation consists of helium nuclei, each composed of two neutrons and two protons. We have already discussed when an emission of alpha radiation may be expected to occur. Approximately, from the element zinc upwards, the binding energy per particle decreases with the increase of the number of particles, due to the increasing intensity of the electrostatic forces of repulsion, and consequently, from the point of view of energetics, the emission of an alpha particle by the heavier elements may be advantageous under certain circumstances. When this occurs, the energy and range of the alpha particle are determined by the difference in mass defect between the original nucleus and the nucleus which is the product of the process. Thus, all alpha particles which are the product of a specific decay process have the same range (Figure 3). Actually, most of these alpha emitters are at the end of the periodic table of elements. Radium and uranium are the best-known ones.

One might be inclined to assume that a nucleus in which the principles of energetics admit the possibility of the occurrence of alpha decay, would disintegrate immediately or at least within a very short period of time. However, that this is not the case is proved by the fact that large quantities of